Teleoperation Receiver Dongle Mounting Height

Standard Operating Procedure · KPR-SOP-001 · Rev A

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Purpose

This document specifies the required mounting height for teleoperation receiver dongles on conductive robot chassis, and the physical basis for that requirement.

Scope

This procedure applies to Bluetooth receiver dongles mounted on conductive robot chassis during teleoperation at working ranges up to approximately 2 m. It supersedes prior guidance to tape the dongle directly to the chassis.

Background

During teleoperation, a continuous two-way RF (radio frequency) link between the operator's handheld controller and the robot is necessary for safe and smooth operation. Signal dropouts can lead to unpredictable movements or run-stops (emergency stops), which can cause the robot to stall in an unsafe position or to accelerate unexpectedly. Unexpected motion also degrades data quality, which hinders training of ML (machine learning) models.

The controllers transmit over Bluetooth in the 2.4 GHz ISM band. Receiver dongles are attached at the center of the robot base with a 0.3 m USB 3.0 cable. Prior to this document, operators were instructed to tape the receiver dongle directly to the top of the chassis base. A research and development technician (Kevin P. Rodriguez) noted anecdotally that dropouts were less frequent when several centimeters of space were added between the chassis surface and the dongle.

Problem Statement

Existing guidelines for teleoperation instruct operators to tape Bluetooth receiver dongles flat onto the top surface of the conducting chassis. The receiver then sits at or near an EM (electromagnetic) field node, thereby weakening signal strength and increasing the frequency of dropped packets.

Analysis

A freely propagating EM wave incident upon a conducting surface will generate a reflected EM wave of the same frequency traveling in the opposite direction. In the absence of other media, the incident and reflected wave will interfere destructively at certain points in space (weakening

signal), while interfering constructively at others (boosting signal). When the speed of the EM wave is also known, these points in space can be calculated explicitly.

During operation, the RF signal hops across the 2.400–2.4835 GHz ISM band hundreds of times per second. Taking the propagation speed as $c \approx 3.0 \times 10^8$ m/s, the free-space wavelength therefore varies between $\lambda = c/f \approx 12.1$ cm and 12.5 cm across the band. (Air's index of refraction at STP departs from unity by roughly 0.03% and is neglected.)

Assuming the chassis to be a perfect conductor to first order, the electric field is required by Maxwell's equations to vanish inside the conductor. This in turn requires the tangential component of the electric field due to both the incident and reflected EM waves to similarly vanish at the surface of the conductor, which forces the latter to acquire a 180° phase reversal. For an EM wave with electric field strength E_0, the magnitude of the spatial envelope is found to be proportional to $2E_0|\sin(kz)|$, where $k = 2\pi/\lambda$ and z is the distance from the chassis surface.

As shown in Figure 1, the resulting envelope of the combined field amplitude of the incident and reflected waves is minimized at the nodes (integer multiples of $\lambda/2$) and maximized at the antinodes (odd multiples of $\lambda/4$). For $\lambda = 12.1$ cm and 12.5 cm, $\lambda/4$ lies between approximately 3.0 cm and 3.1 cm. The envelope at the band center (2.442 GHz, solid line) departs only slightly from the band-edge envelopes (2.400–2.4835 GHz, shaded) out to the first antinode.

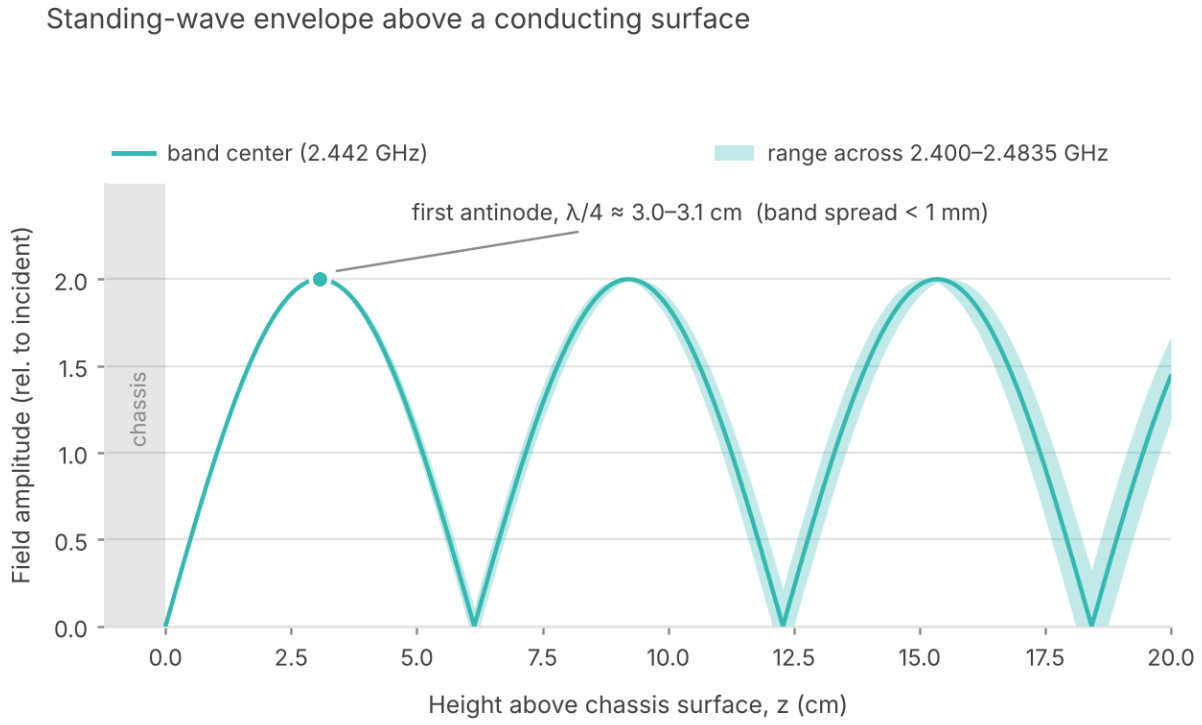


Figure 1: Standing-wave envelope above a conducting surface at normal incidence, across the Bluetooth band.

Figure 2 shows the resulting benefit of the λ/4 offset as the transmitter moves away from the robot, using a two-ray model with the transmitter 1.22 m above the chassis plane. The full +6 dB gain holds near normal incidence and is maintained to about 0.5 m of horizontal separation; the gain remains at least +3 dB out to about 2 m and falls to zero (break-even) near 3.5 m. The 2 m working range stated in Scope follows from this curve.

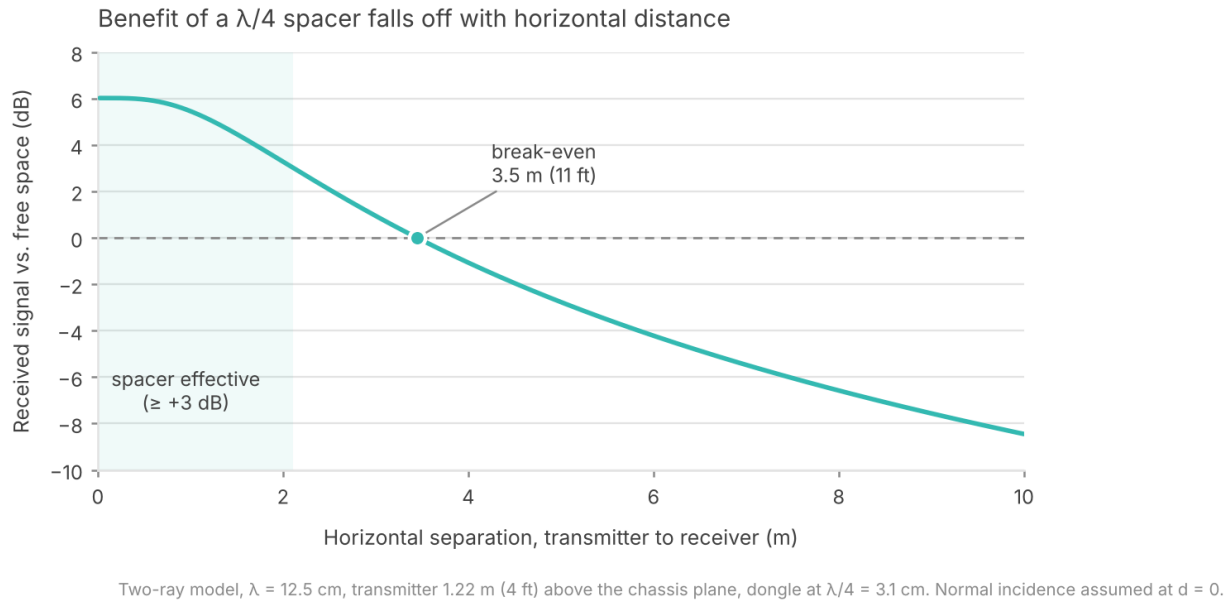


Figure 2: Benefit of the λ/4 offset as a function of horizontal transmitter–receiver separation.

Recommended Procedure

The receiver dongle shall be mounted 31 ± 1 mm above the chassis surface, at the chassis center, using the printed mount described below. Where a printed mount is not available, the interim method may be used until one is.

Materials

- Receiver dongle with 0.3 m USB 3.0 cable
- Printed dongle mount, part number MNT-001, nominal height 31 mm
- *Interim method only*: non-conductive spacer, 20–40 mm tall (cardboard, foam, plastic; ***not*** metal)
- Tape for chassis surface

Mounting with the printed mount (preferred)

1. Confirm the mount is the correct part and is undamaged. If the mount height is in doubt, measure it: 30–32 mm from base to dongle seat.
2. Position the mount at the center of the chassis top surface, at the location previously used for direct taping.
3. Secure the mount to the chassis with tape.
4. Seat the dongle in the mount with its long axis parallel to the chassis surface. Confirm the dongle does not contact the chassis at any point.
5. Route the USB cable to the host port. Secure the cable so that it does not load the dongle or pull it out of the mount during operation.
6. Proceed to Verification.

Interim method (no printed mount available)

1. Select a non-conductive spacer 20–40 mm tall. Do not use a metallic object or anything with a metallic surface.
2. Place the spacer at the center of the chassis top surface and secure it to the chassis.
3. Tape the dongle to the top of the spacer. Confirm the dongle does not contact the chassis.
4. Route and secure the cable as in step 5 above.
5. Replace the spacer with a printed mount at the first opportunity.
6. Proceed to Verification.

If dropouts persist after mounting

1. Confirm the dongle is seated at the specified height and is not in contact with the chassis.
2. Confirm the transmitter is within 2 m horizontal distance of the robot during operation. Beyond approximately 3.5 m the mounting offset provides no benefit (see Limitations).
3. Do not increase the mounting height beyond 40 mm. A height near 6 cm places the dongle at the next field node, where it is no better off than when taped flat, and no taller spacer helps within the working range.
4. If dropouts persist with the above confirmed, report to Operations Lead for antenna evaluation. Do not continue adjusting the mount.

Verification

Acceptance check (each installation)

After mounting, run a 5-minute teleoperation session at normal working range and note any stalls or run-stops attributable to signal loss. Zero events constitutes acceptance. If events occur, follow *If dropouts persist* above.

Validation record

A ~ 30 mm offset using stacked non-conductive spacers was tested on eight units in December 2025. All eight showed a marked reduction in dropout frequency relative to direct chassis mounting. Printed mounts at approximately 30 mm were subsequently produced and were in field use as of January 2026. No instrumented validation of the printed mount has been performed. See Limitations and Follow-On.

Limitations

- Normal-incidence model; oblique incidence moves the optimum up, never down
- ≤ 2 m working range; break-even ~ 3.5 m
- Near-field effects not modeled: 3 cm is only $\sim 1.6 \times$ the reactive near-field radius ($\lambda/2\pi \approx 2$ cm), so the chassis loads the antenna as well as reflecting from it
- Perfect, infinite conductor assumed; real chassis is neither
- Validation was n = 8, qualitative, and uninstrumented
- The printed mount has not been validated; all testing to date used interim spacers at approximately 30 mm

Follow-On

- Instrumented measurement: packet loss and latency distribution, before/after, across ≥20 units, height swept 0–50 mm
- Release the printed mount as a controlled part with drawing and ID
- Evaluate antenna relocation for operations beyond 2 m